Higher National Diploma in Information Technology

First Year, Second Semester Examination – 2023

HNDIT 2022 - Software Development

Instructions for candidates:
Answer **five** questions **only.**
All questions carry equal marks.

No. of questions : 06
No. of pages : 04
Time : 03 hours

Question 01 **[Total marks 20]**

i. Define the term 'computer program'. (03 Marks)

ii. List three (03) types of software. (03 Marks)

iii. Fill in the blanks using most suitable word from the given list of words. (04 Marks)

(Selection, Data structures, String, Integrated Development Environment)

a) _____ is a data type used to assign textual data.

b) _____ are important in computer programming for organizing, managing, and storing data quickly and efficiently.

c) Netbeans is a popular _____.

d) _____ is one of three basic types of control structures.

iv. Briefly explain two (02) characteristics of an algorithm. (04 Marks)

v. Create an algorithm that multiplies two numbers and displays the output. (06 Marks)

Question 02 **[Total marks 20]**

i. What are the three (03) development techniques? (03 marks)

ii. Read the following case studies and mention which development technique [from the techniques you mentioned in the above part (i)] is used in each case study. (03 marks)

a. Case study 1:

Nimal is creating a program for a small library. First, he writes one part of the program to add new books. Then, he writes another part to search for books. Finally, he writes a part to print the list of available books. Each part is written as a separate section and tested on its own.

Which development technique is Nimal using?

b. Case study 2:

Amaya is building a program to create new user accounts. She checks if the user leaves the name or email empty. If there is a mistake, the program shows a clear message instead of stopping. It also checks if the password is strong enough.
What development technique is Amaya using to prevent errors?

c. Case study 3:

Ravi is writing a program to find the total number of files in a folder. Inside certain folders, there are more folders. His program goes into each folder, checks for more folders, and does this again and again until no more folders are left.
Which development technique is Ravi using to solve this problem?

iii. SDLC is the process of building software from the idea stage to the finished product, and even after it is used by people. Briefly explain any four (04) stages of SDLC. (04 marks)

iv. Identify four (04) defects on the following user registration interface. (04 marks)

v. The process of checking the functionality of an application as per the customer needs without taking any help of automation tools is known as manual testing. Explain the three (03) types of manual software testing. (06 marks)

Question 03 **[Total marks 20]**

i. Define the term 'Data Structure' (03 marks)

ii. State two (02) Linear Data structures and one (01) Non-Linear data structure. (03 marks)
array , stack trees

iii. All data in the data structures are processed by some specific operations .Name 04 basic operations that can be performed on data structures. (04 marks)

iv. Define the term 'Sorting', and state two (02) sorting algorithms. (04 marks)

v. Arrange the list shown below in ascending order using insertion sort algorithm. (It is required to show the steps or passes). (06 marks)

18, 10, 12, 4, 5

Question 04 **[Total marks 20]**

i. Why is 'documentation' important? (03 marks)

ii. State three (03) major types/styles of documentations used in Software development. (03 marks)

iii. Define the term 'Security' in Software development, and mention two (02) importance of software Security. (04 marks)

iv. Briefly explain any two (02) of CIA attributes in software security? (04 marks)

v. a) Briefly explain the difference between 'Primitive Data types' and 'Derived data types' with examples. (02 marks)

b) In most structured programs, we can divide the code into three main parts:

- Statements to establish the start of the program
- Variable declaration
- Program statements

Look at the following program and list the correct lines of code under above each parts. (04 marks)

```
public class Hello {
    public static void main(String[] args) {
        int number = 10;
        String name = "Nimal";
        System.out.println("Hello, " + name);
        number = number + 5;
        System.out.println("New number is: " + number);
    }
}
```

Question 05 **[Total marks 20]**

i. File handling refers to the process of performing operations on files. Mention three (03) operations that can be performed on a file. (03 marks)

ii. Define the term 'Stream', and mention two (02) types of streams used in Java to perform file I/O (Input/Output) operations. (03 marks)

iii. Briefly explain the difference between UI design and UX design. (04 marks)

iv. Algorithm analysis helps determine the efficiency of an algorithm based on the number of steps it takes to complete a task. The three common cases used in analysis are Best Case, Average Case, and Worst Case.

Read the following scenario carefully and fill in the blanks with the correct case. (04 marks)

Scenario:

A librarian is searching for a specific book on a shelf that holds 100 books, are not in order. The librarian checks books one by one from left to right.

a) The librarian finds the book after checking about 50 books. This is the _____ case of searching.

b) The librarian finds the required book as the last book on the shelf. This is the _____ case of searching.

c) The librarian finds the required book as the first book on the shelf. This is the _____ case of searching.

d) The librarian checks every book but still doesn't find the required book. This is the _____ case.

v. Read the following case study and answer the question. (06 marks)

A school canteen has introduced a digital ordering system where students can order their meals using a touchscreen. The system displays a menu of available food items (e.g., rice, sandwich, juice). A student selects an item, confirms the order, and then the system checks whether the item is in stock. If it is available, the system deducts it from inventory, generates a bill, and prints a token for pickup. If the item is not available, a message is displayed saying "Item out of stock".

Question:

Based on the scenario above, write a step-by-step algorithm to describe the process of ordering a meal from the canteen using the touchscreen system.

Question 06

Write short notes on following topics **[04 x 5= Total 20 marks]**

i. System software (04 marks)

ii. Cloud Computing (04 marks)

iii. Object Oriented Programming (04 marks)

iv. Automation testing (04 marks)

v. Debugging (04 marks)